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Toward a neuroscience of natural behavior

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Abstract

One of the most exciting new developments in systems neuroscience is the progress being made toward neurophysiological experiments that move beyond simplified laboratory settings and address the richness of natural behavior. This is enabled by technological advances such as wireless recording in freely moving animals, automated quantification of behavior, and new methods for analyzing large data sets. Beyond new empirical methods and data, however, there is also a need for new theories and concepts to interpret that data. Such theories need to address the particular challenges of natural behavior, which often differ significantly from the scenarios studied in traditional laboratory settings. Here, we discuss some strategies for developing such novel theories and concepts and some example hypotheses being proposed.

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The challenges of studying natural behavior

Systems neuroscience owes much of its success to an empirical tradition of using simplified laboratory settings that carefully control sensory stimuli, motor responses, or other task-relevant variables in order to identify the neural mechanisms underlying different behaviors. However, while this approach facilitates the interpretation of neural signals, it potentially limits our theories to explain only those highly simplified scenarios without addressing the true complexity of natural behavior. For this reason, many in the field are now seeking to move beyond simplified laboratory settings toward more naturalistic scenarios in contexts such as navigation, foraging, and even fully unconstrained behavior in open environments [1-10,11*,12*,13**,14**]. This goal is not new, of course [15,16], but only recently have new technological advances made it possible to widely adapt it to the particular demands of empirical neuroscience. These advances include the capacity for wireless recording in freely moving animals [17–19] and automated systems for quantifying behavior [20–23], as well as new mathematical methods for analyzing the large datasets produced by such methods [24–27]. As a result, we are now seeing an increasing trend for studies that go beyond the tradition of constrained laboratory experiments and move toward addressing more naturalistic behavior.

This important trend, however, comes with its own challenges. A particularly serious one concerns interpretability. As we move away from the constraints of simplified laboratory settings, we also move away from many of the approaches we have relied upon to make sense of neural data, which is highly complex and variable. Classic studies succeeded in interpreting neural data by independently controlling theoretical variables of interest (e.g., visual motion [28], direction of hand movement [29], whole-body motion [30]) to show how they influence neural activity. But in a freely moving animal, these variables may not be systematically sampled and/or often covary, making them challenging to dissociate. This makes interpretation difficult because it is often impossible to know whether a given pattern of neural activity is related to a sensory variable that informs the guidance of movement, to the descending commands that produce the movement itself, or to other, unknown factors that covary with one or both [31**].

One strategy is to use a "data driven" approach by collecting large sets of data and then using sophisticated analyses to extract patterns of covariation or functional clusters [26,27]. While this is informative in principle, the implications it has for understanding neural mechanisms are limited by two key problems. First, as noted above, freely moving animals behave in ways that don't necessarily lend themselves to dissociating between different sensory, motor, or cognitive variables which are changing simultaneously and in a correlated fashion. Consequently, the ability to decode a particular variable from the activity pattern across a given neural population does not necessarily imply that the function of the population has anything to do with that variable. Even

sophisticated methods to regress out potential confounds cannot distinguish between variables that, for good functional reasons, are dynamically coupled. For example, recent studies have shown that spontaneous movements strongly influence neural activity across a variety of cortical areas in mice [32–34] and monkeys [31**]. This has even been shown for primary visual areas [33,34], raising the possibility that under natural conditions motor control is such an intimate aspect of both sensory sampling and cognitive processes that there are no clear borders between these functions within cortical circuits. On the other hand, a recent study in monkeys showed that what might appear to be movement-related activity in the visual cortex can be almost entirely explained by the visual input changes caused by gaze shifts [35*]. Collectively, such studies demonstrate the importance of careful measurement of both movements and changes in sensory stimulation, no matter how unrelated to the task they may seem [31**].

The second limitation of decoding-based approaches is that they are primarily descriptive and not designed to distinguish between different hypothetical mechanisms responsible for neural activity. Almost any sensible model will predict some relationship between neural activity and task-relevant variables, so the existence of such a relationship does not provide sufficient constraints to distinguish between different candidate mechanisms. Furthermore, comparisons of decoding accuracy from different brain regions may not necessarily be conclusive because the results of such analyses will often be dominated by differences in signal-to-noise ratios, electrode placement, or potentially flawed assumptions about the relevant variables to decode (e.g., movement trajectory, muscle force, or a suitable command signal to a downstream system with its own intrinsic dynamics). Consequently, purely data-driven methods typically cannot reveal functional principles even when applied to a well-understood and highly structured system [36]. It has been argued that "coding" is really a way of describing brain activity in a way that is independent of the underlying causal mechanisms and for this reason it cannot serve as a path for understanding them [37]. In fact, even sophisticated procedures that fit neural data using deep learning cannot in general identify the mechanisms actually used by the brain, because there are always many ways a given problem can be solved [38,39**]. To understand underlying mechanisms, there is no substitute for defining specific hypotheses that describe them and conducting targeted experiments that can distinguish between the predictions of candidate hypotheses [40*].

Consequently, to understand the neural mechanisms of natural behavior we need not only to develop better methods for neural recording, behavioral quantification, and data analysis, but also to revise our theoretical frameworks to better address the particular challenges actually faced by animals interacting with their natural environment [9,41]. As noted at the outset, many of the theories upon which our interpretation of neural data is based are inherited from the results of studies that used highly simplified experimental tasks. Thus, those interpretations may embed assumptions that simply do not hold during many aspects of natural behavior. This raises the concern that the theories built on the basis of such studies may not generalize to natural behavior and will require significant revision.

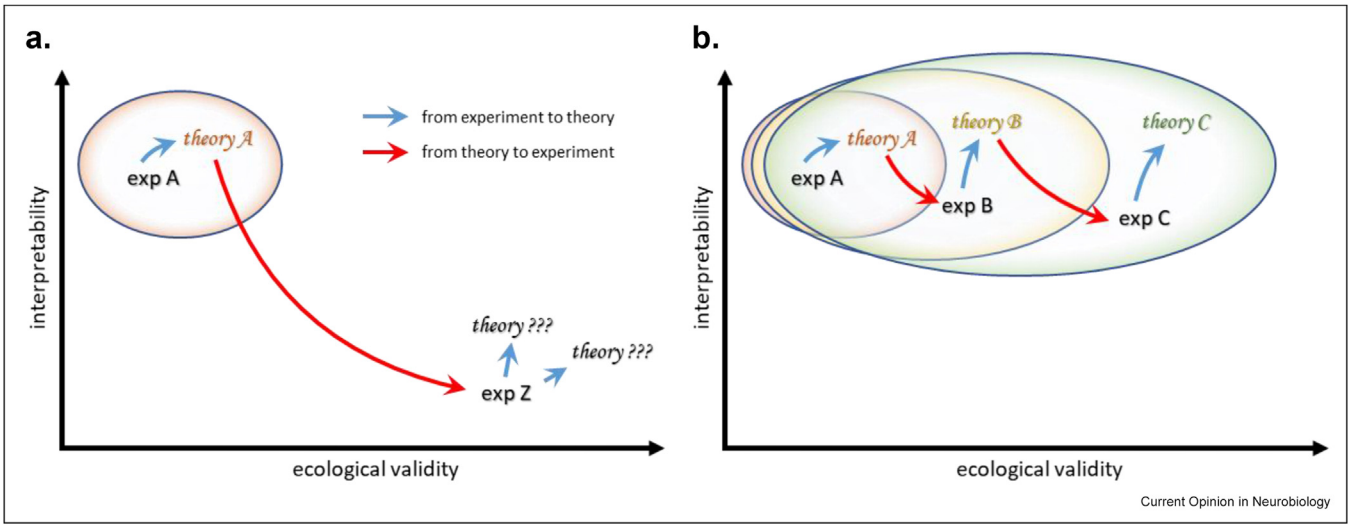
Moving neuroscience toward the natural world

But how can we move away from simplified settings and toward a neuroscience of natural behavior, in both an empirical and theoretical sense, without sacrificing our ability to interpret data in the context of specific mechanistic hypotheses and without losing what has already been learned? Jumping directly to completely freely moving animals risks producing data that lie far outside of the scope of current theories (Figure 1a), limiting us to mainly descriptive analyses. An alternative strategy is to instead take relatively modest steps, each gradually capturing additional aspects of natural behavior while retaining precise experimental control and data interpretability (Figure 1b). For example, many laboratories have begun to use virtual reality paradigms that present humans or other animals with scenarios that mimic many aspects of natural behavior but nevertheless allow full control over the stimuli and how they are modified by the animal's behavior [7,42-46,47*,48*].

Moving toward natural behavior in more modest steps makes it possible to retain our ability to interpret data even as we gradually release experimental constraints. Key to this is the ability to directly compare neural activity in both constrained and more unconstrained settings, letting the former guide the interpretation of data from the latter. Ideally, one could make such comparisons even at the level of individual neurons, making use of chronically implanted recording systems [49,50]. However, here too there are challenges. In particular, as we gradually move toward more naturalistic scenarios, we want to ensure that our new theories are still compatible with data under the more restricted conditions in which our original ones were developed. At the same time, we do not want to overly limit ourselves from new theories that have the breadth to explain the more natural scenario. How can we go about achieving that somewhat difficult balance?

The first step is to clearly identify which of the assumptions built into our current theories based on more restricted studies no longer hold under more natural scenarios. Then we can design a less restricted experiment in which one or a few of those assumptions are relaxed and examine whether our data does or does not remain consistent with our previous theory. If it does,

Figure 1



The challenge of studying natural behavior is that by increasing ecological validity, we risk losing the interpretability of our data. **a.** Jumping from highly constrained and unnatural settings (exp A) all the way to fully freely moving scenarios (exp Z) produces data that is difficult to interpret because it lies far outside of the range of assumptions or constraints upon which the existing theory (theory A) was built, and thus beyond the scope of observations (red oval) that this theory can explain. It is therefore difficult to know how to modify the theory to make the new data interpretable. **b**. Making smaller steps by gradually releasing the constraints of our experiments (exp B) makes it more likely that we will find a way to extend or modify current theories to broader ones (theory B), which reflect more ecologically valid assumptions and can explain the new data as well as the old. That process can then continue to new experiments (exp C) and new theories (theory C) that have even higher ecological validity without losing interpretability along the way. Note that ovals in the figure represent the range of experimental observations that can be explained by a given theory but do not necessarily imply overlap between the theories themselves and/or the assumptions embedded within them.

then the assumptions inherent in our interpretation of the data were not crucial to the success of the theory, which can now be generalized to the less restricted scenario. If it does not, then we need to carefully step back to examine which aspects of our theory were dependent on those flawed assumptions and then broaden or change it to explain the data obtained under the new scenario.

This kind of approach has already yielded important insights in studies that progressively moved from simple to more complex actions involving multiple body segments. For example, classical studies of the superior colliculus (SC) in head-fixed monkeys revealed topographic mappings and temporal correlations between neural activity in deeper layers of the SC and the amplitude and direction of saccades, leading to the notion that the SC encodes and controls saccadic eye movements [51–53]. However, when the effects of stimulation and neural activity were reexamined in a slightly more natural scenario, in head-free monkeys and cats, it became clear that in fact these same neural populations actually drive and encode gaze shifts achieved through a combination of eye and head movements [54–57]. Importantly, these findings did not completely invalidate the previous theory of SC function but rather required its interpretation to be extended to more natural, head-free behavior through carefully designed experiments and analyses which allowed dissociation of eye, head, and gaze (i.e., combination of eye and head) variables [55,58–60]. Other studies have used richer sensory stimuli and tasks to further extend theories of SC contributions to sensorimotor control to incorporate cognition [61]. Importantly, such findings are consistent with studies in a wide range of species suggesting that the SC (optic tectum in non-mammals) is more generally involved in integrating multimodal cues to select and guide actions of varying complexity, of which eye movements are just one component, including whole-body orienting and locomotion for the purposes of approach and avoidance/escape [59,62-65,66*]. Thus, although theories of SC function based on studies in primates have emphasized its role in vision and gaze control, investigation of increasingly naturalistic behavior may ultimately reveal much broader functionality. Indeed, this is supported by a number studies of providing evidence for a role for deep layers of the primate SC (albeit not necessarily the same neural populations as those involved in gaze control) in defensive responses [67], as well as in arm movements [59,66*,68-72] that potentially mediate rapidly initiated or "express" visuomotor responses [73–76] and/or eye-head-hand coordination [77]. More generally, our theories for the sensorimotor processing

underlying the coordination of actions involving multiple body segments within sub-cortical and cortical networks continue to evolve as we gradually move away from the more restricted scenarios in which motor behaviors have traditionally been studied (e.g., reaching in seated, stationary subjects [29,78–80]) toward the less-restricted scenarios we encounter in daily life (e.g., reaching as the body moves through space [81-84,85*]).

Another example arises in studies of decision-making. Classically, decision-making is defined as a cognitive process that is separate from sensorimotor control. From that perspective, it makes perfect sense to study decision-making using paradigms in which the agent is presented with some options, makes his/her choice, and then later reports that choice with a voluntary movement such as a button press, reaching action, or saccade. Such "decide-then-act" paradigms allow one to distinguish a process of deliberation, usually about a static set of options, from the commitment to a choice and the preparation and execution of a movement. However, while this makes intuitive sense for the kinds of abstract decisions that dominate human lives, it is not well suited for many of the kinds of decisions that animals make during natural behavior. The reason is that during natural behavior actions and decisions often unfold simultaneously. Consider a lioness chasing antelopes in a herd or a soccer player running among opponents toward a goal. In such scenarios, each agent is continuously engaged in performing an action and adjusting it through feedback, while remaining sensitive to new potential options that may present themselves, as objects in the world and their relationships to the agent continuously change. Such "decide-while-acting" scenarios [86,87,88*,89*,90*] violate many assumptions of classical experimental paradigms and theoretical models. For example, they present a challenge to theories that describe the transition from deliberation to commitment as the crossing of a "decision threshold" [91–93], because whatever neural circuit is responsible for ongoing action must already be past its threshold. For such scenarios, there is more promise in models in which decisions and actions are part of a high-dimensional unified system [94*] in which deliberation about alternative actions can take place within a subspace of neural activity that controls the current action without influencing its execution [88*,95]. This calls for an elaboration of current theories beyond the domain of the tasks they were originally developed to explain.

Developing theories for natural behavior

If a previously developed theory fails under less restricted conditions, how do we go about modifying or changing it to be better suited to natural behavior? Here we are faced with deciding to what extent we can retain valuable components of previous theories without limiting ourselves from embracing new conceptual ideas about function and underlying mechanisms. Deciding on how to achieve that delicate balance will benefit from tracing back the historical origins of the existing theory to the point where the assumptions that have now been shown to be flawed were built into it. This involves clearly identifying both the experimental observations that were made as well as the assumptions that led to their interpretation in support of that theory. In general, this is a very difficult task because those assumptions are often unstated [96], and some are embedded in the very language we use to describe the topics of our research [97,98]. For example, "cognition" is classically defined to include processes like thinking or reasoning but not mechanisms of sensory processing or motor control [99,100]. Under that assumption, studies examining neural activity in sensory or motor regions do not shed light on cognition. However, not only have decision variables been found throughout "sensorimotor" areas [61,101–107] but motor variables are found in "cognitive areas" [12*,31**,33,108], casting doubt on the validity of that assumption. Recent work instead shows a continuous gradient between motor-related versus choice-related activity across the frontal cortex [12*], consistent with the idea that prefrontal areas are an extension of premotor mechanisms toward more abstract domains [109,110], and with a long history of proposals that cognition is embodied and an extension of sensorimotor control mechanisms [111–120].

Having identified the assumptions we need to abandon or modify, we need to propose alternative hypotheses to explain both the data upon which the original (incorrect or incomplete) theory was built, as well as our new observations obtained under less constrained conditions. We are also faced with the challenge of designing experiments that are flexible enough to address our new hypotheses under those less restricted conditions while still maintaining sufficient control to test their validity. For example, hypotheses which aim to explain the apparent mixture of motor and cognitive variables across a broad range of brain regions will ultimately need to be tested through experiments that go beyond traditional attempts to dissociate them (e.g., by keeping one or the other set of variables constant or temporally separated), because mixing of such variables may in fact be the key to how the system works. If that's the case then we need experiments that are explicitly designed to examine their simultaneous interaction under natural but controlled conditions (e.g., how spatial relationships among targets interact with decision variables [104], and how movement influences traditionally cognitive functions such as decisions [84,88*,89*,121-125], memory retrieval [126], semantic processing [120], goal tracking [46], etc.).

To develop these new theories and experiments we can draw inspiration from knowledge on how the species we study actually behave in their natural habitat, benefitting from a long history of work in ethology and

neuroethology [9,127–131]. One example of how ethology potentially leads to novel frameworks arises in studies of economic decision-making. Many studies of decision-making have been inspired by the kinds of scenarios humans face in their daily lives, such as when selecting a meal from a restaurant menu [132]. In such scenarios, options are presented simultaneously and one can choose between them by comparing some estimate of their subjective value in terms of quantity, taste, cost, etc. This leads to hypotheses in which the brain calculates the value of each option, compares them, and then prepares an action [133]. However, most animals in the wild rarely encounter simultaneous food items. Instead, they tend to come across a single food item and must decide whether it is worth obtaining or whether time and energy are better spent looking for something better [134,135]. This suggests an alternative hypothesis on the neural mechanisms of decisions, whereby there is a competition between neural systems involved in accepting and obtaining an available offer versus other systems involved in motivating one to search elsewhere [131,136–140].

Another source of insights can come from the field of ecological psychology, which emphasizes the close integration of sensory and motor functions [141]. As noted above, behavior in a psychological experiment can be described as a serial process of perceiving the world, thinking about it, and finally acting on it, but such a serial process cannot be effective during natural behavior, in which animals are continuously moving through and interacting with a dynamically changing world. In particular, at any given time, the action choices available to us are defined by the spatial relationships of objects around us, our current and future relationship to those objects, and the types of physical constraints and costs (e.g., spatial and biomechanical) inherent in a given action choice. For example, some objects specify actions precisely (e.g., reaching for the handle on a coffee cup) while others specify a broader range of potential trajectories (e.g., walking through a wide doorway). Some spatial layouts force decisions between distinct families of actions (e.g., reaching around an obstacle) that are influenced by biomechanical constraints (e.g., which way around the obstacle is less effortful). All of these relationships are continuously changing as we move our bodies through the world, and new opportunities and decisions emerge. Such scenarios require theoretical frameworks that emphasize the "closed loop" nature of behavior, in which sensory information is used to specify a "landscape" of potential actions [142–144], which is continuously updated as we move through and interact with the world, and modulated by action costs, predicted outcomes, and future actions that may become available [41,144,145]. Importantly, all of these processes must take place in parallel, even during ongoing action, which can itself influence decision processes [84,88*,89*,121-125]. This necessary integration may be one reason why even putatively perceptual decisions can be biased by the subtle costs of the movements used to report them [146,147], why movement can influence cognitive processes [120,126], and why movement variables are so widely distributed in brain activity [3,32,33].

A complementary strategy for developing theories that address natural behavior is to take an evolutionary perspective [148,149,150**]. The mechanisms we find today in modern animals result from a long process of evolutionary changes, each adapted to the context of a particular niche that our ancestors occupied at a particular stage in history. Importantly, because evolution always works from an existing ancestral system, many of the early mechanisms and circuits still exist in modern animals and form the building blocks for their behaviors [151**]. Thus, reconstructing the evolutionary sequence can tell us a lot about how specific circuits elaborated and specialized along a given lineage of interest. In particular, it can significantly reduce the space of candidate hypotheses toward those that are most biologically plausible and potentially shed light on functional concepts better suited to understanding natural behavior.

For example, Cisek [149,150**] describes a putative evolutionary sequence of how the brain was gradually elaborated along the lineage that produced primates. It begins by considering the relatively simple closed-loop behavior of our distant ancestors and then, guided by comparative, developmental, and fossil data, proposes how more complex behavior and its underlying neural circuits were progressively elaborated over evolutionary time. This yields a functional architecture that differs from classical frameworks, but is claimed to be more compatible with neural data. In particular, it does not include categories such as "cognition", of which "decision-making" is a sub-category, whose neural substrates are expected to be localized in specifically "cognitive regions". Instead, it suggests that the functional architecture consists of parallel sensorimotor circuits, each adapted to the needs of particular species-typical natural behaviors [152,153**], whose activation is orchestrated through a hierarchy of selection mechanisms including the hypothalamus, basal ganglia, and ventrolateral allocortex [150**,154]. For example, behaviors involving locomotion, such as searching for food or a place to hide, may involve medial regions including retrosplenial cortex, cingulate, and supplementary motor areas [155–160], while behaviors involving handling and ingesting food involve lateral regions including premotor, motor, somatosensory, and intraparietal cortex [152,161–166].

In other words, from an evolutionary perspective the subdivisions of neural function are not distinct computational modules, but circuits adapted for controlling

ethologically relevant classes of natural behavior [167]. Consequently, understanding the brain will require understanding natural behavior and its specific, pragmatic needs [168] and may also require us to move away from some strongly held assumptions and approaches. For example, many theories treat the brain as an information-processing system that transforms input into output. However, as noted above, during natural behavior animals are moving through and interacting with their environment in a "closed loop" fashion, whereby actions continuously influence sensory information. This reflects the fact that the brain is not really an organ for producing an output in response to sensory input, but a control system whose fundamental role is to control the animal's state (both internal and external) via output through the environment [149,169–175,176*]. Carefully designed experiments that investigate such closed-loop interaction may be among the most valuable steps that will take us toward developing increasingly broad and ecologically valid theories.

Concluding remarks

The endeavor of studying the neural mechanisms of natural behavior presents us with many challenges. Some of these are technological and empirical, and in recent years extraordinary progress has been made along those lines. However, we argue that some of the most difficult challenges facing this endeavor will be theoretical. While studies of constrained scenarios have provided an important conceptual foundation for experimental and theoretical approaches for studying the brain, it is likely that many of the resulting concepts will fail to generalize to explain the richness of natural interactive behavior. Along the way we will need to abandon some deeply held assumptions, perhaps even some foundational concepts [96,149]. However, at the same time, we must not simply discard all that we have learned from classical constrained laboratory studies. Here, we advocate a stepwise strategy for moving toward a neuroscience of natural behavior both in our empirical paradigms and our theoretical frameworks. It involves identifying key theoretical assumptions associated with more constrained conditions, modifying established experimental paradigms in a way that gradually relaxes them, and then explicitly testing whether our current theory holds up in the new situation or whether it must be revised. New theories may benefit from inspiration from multiple fields including neuroethology, ecological psychology, and evolutionary biology. This is a fundamentally hypothesis-driven multidisciplinary strategy, and ideally, one where success is not measured by the degree to which a favorite theory is supported but rather, by the clarity with which alternative hypotheses are discriminated [40]. We believe that ultimately this strategy will facilitate studying increasingly natural behavior without sacrificing the interpretability of our data, allowing us to extend and broaden our theories toward a true understanding of the underlying neural mechanisms.

Declaration of competing interest

The authors declare that no competing interests exist.

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